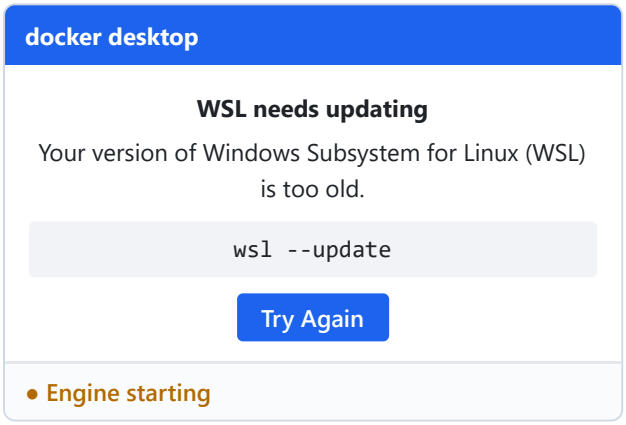
TROUBLESHOOTING GUIDE · DOCKER DESKTOP

Fix: “WSL needs updating”

Docker Desktop won't start on Windows 10 because the WSL 2 kernel is missing or out of date

Project	Spencer Denim E-Commerce Platform	Prepared by	TU2SL
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Applies to	Docker Desktop 4.x · Windows 10 22H2	Related	DEV-GUIDE-01 (Docker setup)

1. Symptom



After installing Docker Desktop and restarting the PC:

- Docker Desktop opens with the message **“WSL needs updating”**.
- The status at the bottom left stays on **“Engine starting”** and never becomes “Engine running”.
- RAM, CPU and Disk show `0.00` or `--`.
- `docker run` commands fail because the engine isn't running.

2. Cause

Windows 10 ships with an old **built-in (“inbox”) WSL**. The command `wsl --install --no-distribution` turns the WSL features on, but on Windows 10 it may **not download the WSL 2 Linux kernel**. Docker Desktop runs its engine inside a WSL 2 virtual machine, so without the kernel it can't start.

The fix is to install the current **WSL package from Microsoft** with `wsl --update`. This is a one-time step.

3. Confirm the cause (optional)

Open a normal PowerShell window and run:

```
wsl --version
wsl --status
```

Command	Shows this problem	After the fix
`wsl --version`	*“Invalid command line option: --version”* (old inbox WSL)	Prints the WSL, kernel and WSLg version numbers
`wsl --status`	*“The WSL 2 kernel file is not found. To update or restore the kernel please run 'wsl --update'.”*	Shows *Default Version: 2* with no kernel warning

4. Fix

1 Quit Docker Desktop DEVELOPER

Right-click the Docker whale icon in the Windows taskbar (bottom-right, near the clock) → **Quit Docker Desktop**.

2 Update WSL ADMINISTRATOR

Open **PowerShell as Administrator** (Start → type *PowerShell* → right-click → *Run as administrator*) and run:

```
wsl --update
```

If it fails, hangs, or reports that the Microsoft Store is unavailable or blocked (common on company PCs), download directly from Microsoft instead:

```
wsl --update --web-download
```

3 Restart WSL DEVELOPER

In any PowerShell window (admin not required):

```
wsl --shutdown
```

4 Start Docker Desktop again DEVELOPER

Open Docker Desktop from the Start menu. If the message still shows, click **Try Again**. Wait until the bottom-left status turns green: **"Engine running"**.

5. Verify

```
# 1) WSL is the current version
wsl --version
WSL version: 2.x.x.x
Kernel version: 6.x.x.x ...

# 2) The Docker engine responds
docker version
#    Must show both a "Client" and a "Server" section

# 3) A test container runs
docker run --rm hello-world
Hello from Docker!
This message shows that your installation appears to be working correctly.
```

> **Fixed:** all three commands succeed and Docker Desktop shows "Engine running". Continue with DEV-GUIDE-01, Part B (first-time configuration).

6. If it still doesn't work

What you see	What to do
`wsl --update` says access is denied or asks for admin	The update must run in **PowerShell as Administrator**. Ask IT if you don't have admin rights.
Both update commands fail (no internet access to Microsoft)	IT downloads the latest `wsl.x.x.x.x64.msi` from the official `github.com/microsoft/WSL/releases` page and installs it as administrator, then restarts the PC.

"Please enable the Virtual Machine Platform Windows feature"	In PowerShell as Administrator:
	`dism /online /enable-feature /featurename:VirtualMachinePlatform /all /norestart`
	`dism /online /enable-feature /featurename:Microsoft-Windows-Subsystem-Linux /all /norestart`
	Then restart the PC and repeat section 4.

"Virtualization support not detected"	IT enables Intel VT-x / AMD-V in the BIOS/UEFI. (Already enabled on the Spencer Denim development PC.)
Engine starts, then stops again	Run `wsl --shutdown`, restart Windows, and open Docker Desktop. Check that `C:\Users\<you>\.wslconfig` (if created) has valid values, e.g. `memory=6GB`.
Message returns after a Windows update	Run `wsl --update` again (as Administrator), then `wsl --shutdown`.

7. Notes for this PC

- **Diagnosis (17 Sep 2026):** WSL features on (Default Version 2), **WSL 2 kernel not found**, `wsl --version` not recognised. Docker Desktop 4.91 and Docker CLI 29.8.0 installed.
- Docker Desktop was installed **for the current user only** (`%LOCALAPPDATA%\Programs\DockerDesktop`). With this install type the `docker-users` group is not used, so DEV-GUIDE-01 Part A step 3 can be skipped.
- Signing in to Docker Desktop (the "Sign in" button) is **not required** to run this project.
- Docker Desktop's built-in Kubernetes is **not used**; leave it disabled.

Document control: DEV-GUIDE-02 v1.0, 17 September 2026, prepared by TU2SL for the Spencer Denim E-Commerce Platform. Source file: `docs/guides/wsl-needs-updating-fix-guide.html`. Related: `Docker-Desktop-Windows-Setup-Guide.pdf` (DEV-GUIDE-01).